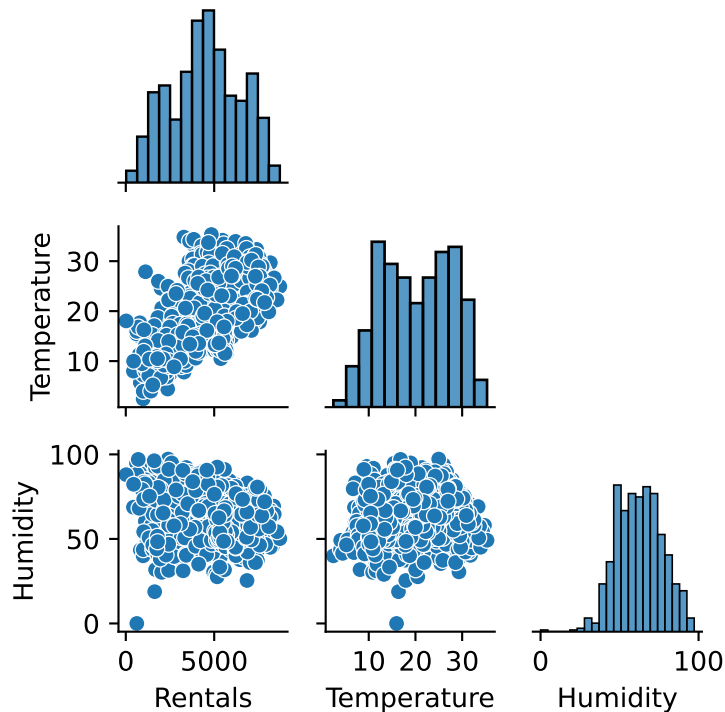


Linear Regression with Bike Sharing Data

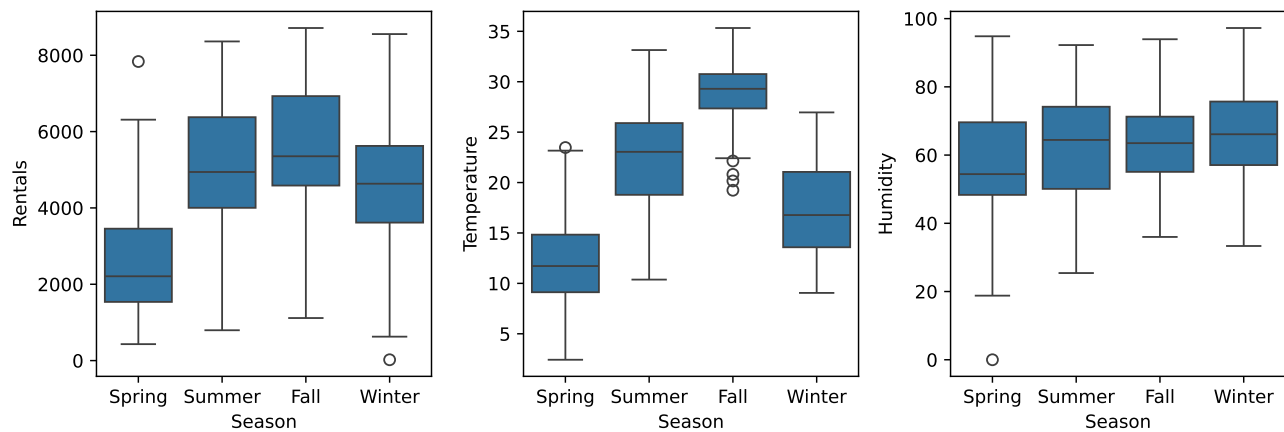
A city operates a bike-sharing system. We want to understand what affects the number of bikes rented on a given day and use regression to predict demand. The dataset contains the following variables:

- **rentals:** total number of bike rentals that day
- **temperature:** temperature in degrees Celsius
- **hum:** humidity in percentage
- **season:** season of the year (spring, summer, fall, winter)

1 Exploratory analysis



Discuss the relationship between the continuous variables in the dataset, two sentences:



Discuss the relationship between the continuous variables in the dataset and the categorical season variable, two sentences:

2 Simple linear regression

We fit a model using only temperature as covariate:

$$\text{rentals} = \beta_0 + \beta_1 \text{temperature} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

OLS Regression Results

```

=====
Dep. Variable:          rentals    R-squared:                0.394
Model:                  OLS       Adj. R-squared:           0.393
Method:                 Least Squares    F-statistic:             473.5
Date:                   Fri, 18 Sep 2026  Prob (F-statistic):      2.81e-81
Time:                   11:14:17    Log-Likelihood:          -6386.8
No. Observations:       731        AIC:                     1.278e+04
Df Residuals:           729        BIC:                     1.279e+04
Df Model:                1
Covariance Type:        nonrobust
=====

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept    1214.6421    161.164      7.537      0.000     898.242    1531.042
temp         161.9685      7.444     21.759      0.000     147.355    176.582
=====

```

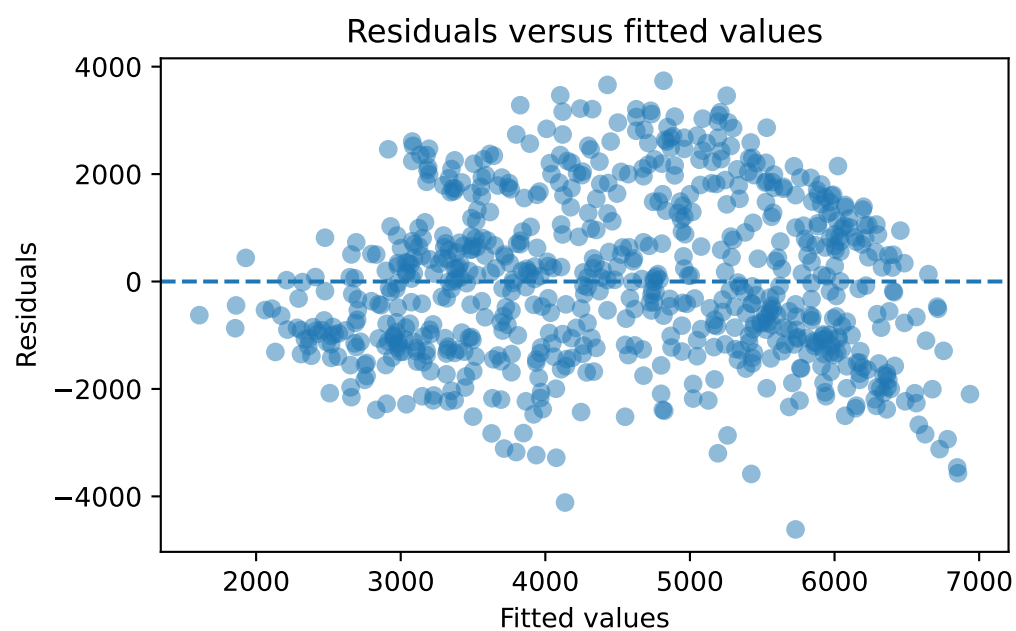
Interpret the temperature coefficient, one sentence:

What does $R^2 = 0.39$ mean? One sentence:

Calculate the predicted number of rentals for a day with temperature 20°C:

Does the small p-value tell us that temperature is important in a practical sense? Discuss the distinction between statistical significance and practical importance. One sentence:

Below you see a plot of the model residuals against fitted values. What does this residual plot suggest about the fitted linear regression model? Discuss and sketch on the figure.



3 Multiple regression

We now include temperature, humidity and season to the model

The fitted model is:

```

=====
                        OLS Regression Results
=====
Dep. Variable:          rentals    R-squared:                0.510
Model:                  OLS       Adj. R-squared:           0.506
Method:                 Least Squares    F-statistic:             150.8
Date:                   Fri, 18 Sep 2026    Prob (F-statistic):      1.11e-109
Time:                   11:14:17    Log-Likelihood:          -6309.1
No. Observations:       731        AIC:                     1.263e+04
Df Residuals:           725        BIC:                     1.266e+04
Df Model:                5
Covariance Type:        nonrobust
=====
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	2500.8267	265.044	9.436	0.000	1980.482	3021.171
C(season)[T.Summer]	874.4701	187.194	4.671	0.000	506.962	1241.978
C(season)[T.Fall]	460.0173	246.006	1.870	0.062	-22.951	942.986
C(season)[T.Winter]	1562.4827	158.231	9.875	0.000	1251.837	1873.128
temp	163.8560	12.072	13.573	0.000	140.155	187.557
hum	-32.5439	3.641	-8.939	0.000	-39.691	-25.396

```

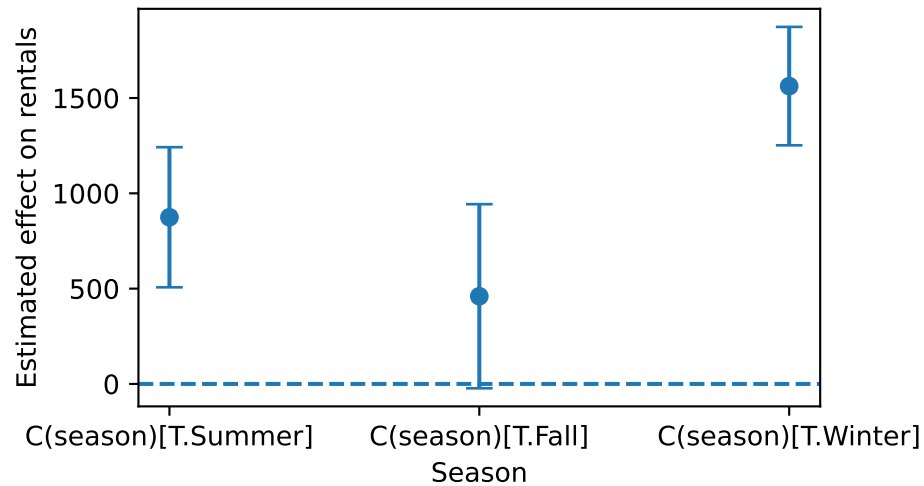
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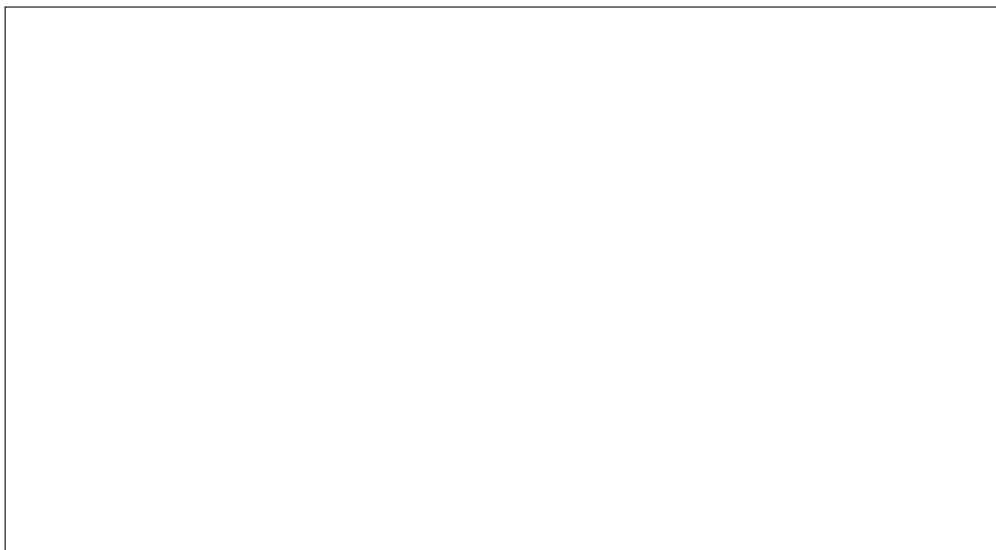
We included season as a categorical covariate in the model. Why this choice? And why in the output there are only 3 parameters for season? Two sentences:

Below are the estimated effect of season. How do you interpret them? One sentence:

Estimated seasonal effects with 95% confidence intervals



Sketch two regression lines, with temperature along the x-axis varying from 0 to 30, and predicted rentals on the y-axis. Humidity should be the same for both lines, but one should be for spring, and the other for winter (you do not need numbers on the figure).



R^2 increased from 0.39 to 0.51 when additional predictors were included. What does this tell us? One sentence:

4 Model with interactions

	coef	std err	t	P> t	[0.025	0.975]
Intercept	1901.4789	352.867	5.389	0.000	1208.712	2594.246
C(season) [T.Summer]	1602.4697	531.562	3.015	0.003	558.878	2646.061
C(season) [T.Fall]	8388.2465	1023.314	8.197	0.000	6379.221	1.04e+04
C(season) [T.Winter]	2023.7786	497.545	4.068	0.000	1046.971	3000.587
hum	-38.5183	3.597	-10.709	0.000	-45.580	-31.457
temp	241.4794	23.202	10.408	0.000	195.928	287.031
temp:C(season) [T.Summer]	-66.6060	30.079	-2.214	0.027	-125.658	-7.554
temp:C(season) [T.Fall]	-317.6347	40.895	-7.767	0.000	-397.922	-237.347
temp:C(season) [T.Winter]	-46.6199	32.152	-1.450	0.147	-109.742	16.502

What is the estimated coefficient of the temperature in Spring and Fall? And what is the estimated intercept for the two seasons? Answers:

Below is a plot that shows estimated regression lines, with confidence and prediction intervals, for Spring and Fall (we have set humidity to zero for simplicity of the task). Which line is for Spring and which one is for Fall?

Explain the difference between prediction and confidence intervals, and why is one type of interval so much wider than the other one? Two sentences:

